# Fieldbus Networks

# Industrial LANs

- Allow interconnection of field devices
  - Sensors
  - Actuators
  - Controllers
- Advantages to point-to-point connections
  - Less wiring
  - Less maintenance
  - Lower cost
- Commonly know as fieldbus networks

# Fieldbus Networks

- Spreading factors
  - Economic aspects
  - Increasing decentralization of control and measurement
  - Increasing use of intelligent microprocessor-based devices
- Network layer structure (H1 level)
  - Physical layer
  - Data link layer
  - Application layer

# Fieldbus Technologies

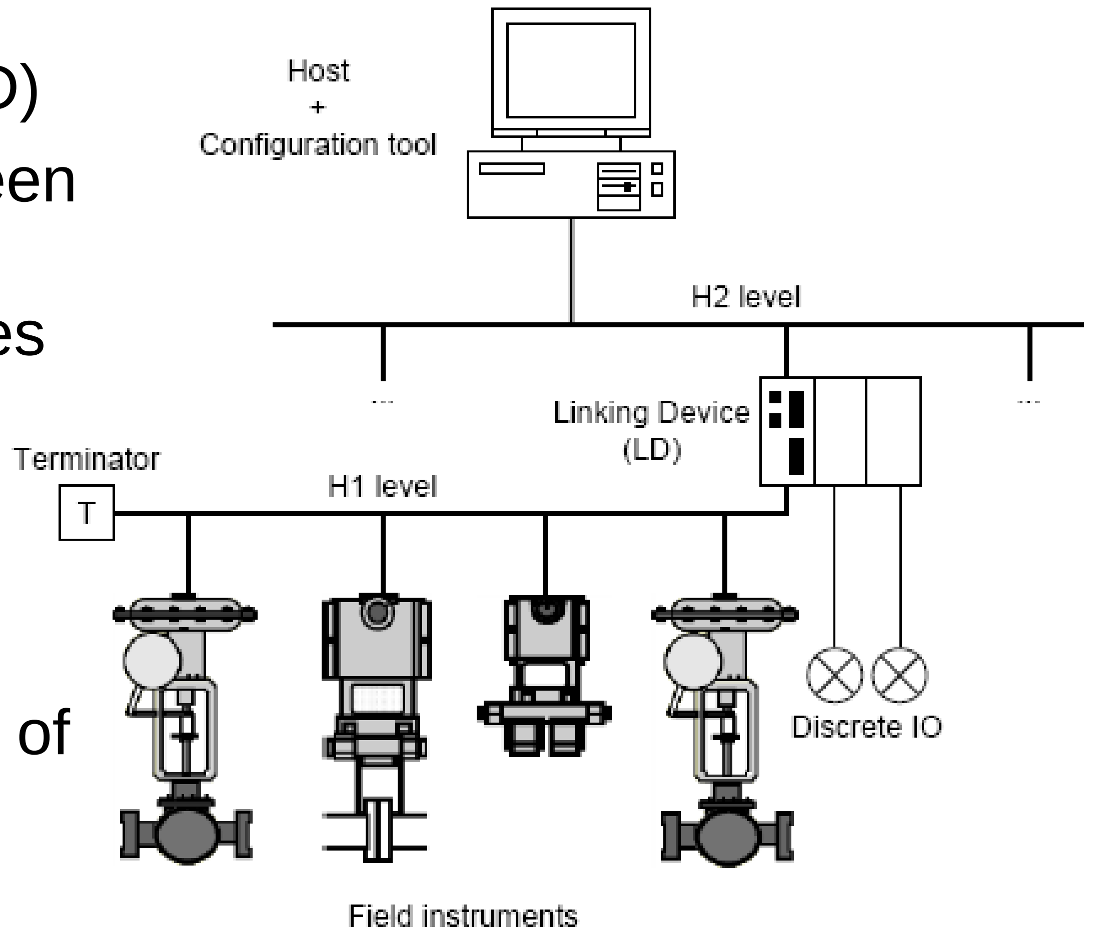
- Fieldbus initiatives started in the beginning of the 80s
  - Factory Instrumentation Protocol (FIP), in France
  - Process Network (P-NET), in Denmark
  - Controller Area Network (CAN), in Germany
  - Process Field Bus (PROFIBUS), in Germany
- Standardization efforts started at international level
  - IEC (International Electromechanical Commission)
  - CENELEC (European Committee for Electrotechnical Standardization)
- Large number of standard protocols, and also proprietary protocols used by control systems vendors

# Fieldbus Levels

- Low Level (H1)
  - Low data rate
  - Interconnection of sensors/actuators in process control
- High Level (H2)
  - High data rate
  - Interconnection of several H1 networks
  - Manufacturing

# Example: Foundation Fieldbus

- Linking Device (LD)
  - Gateway between the H1 and H2 levels (translates the messages exchanged between both levels)
  - Primary master of the H1 level network



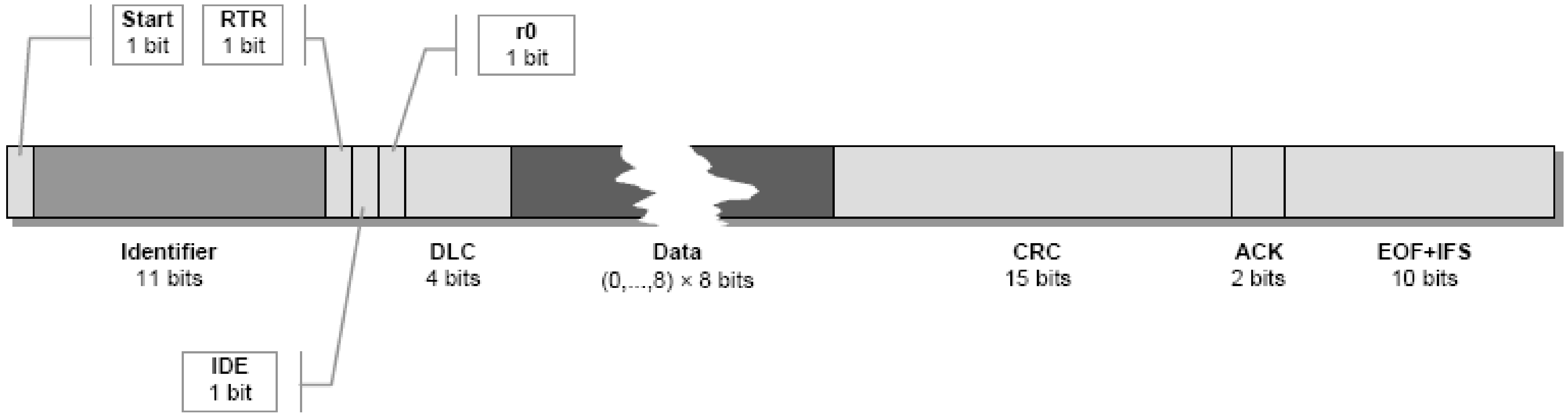
# Fieldbus Requirements

- Efficient handling of very short data blocks
  - Low protocol overhead (packet headers, etc.)
- Ability to handle periodic and aperiodic traffic
  - Periodic – Sample data
  - Aperiodic – Events conditions, such as alarms
- **Bounded delay**
  - Real-time operation
- No single point of failure
  - Some level of redundancy to cover failure of devices
- Low network interface cost

# Controller Area Network (CAN)

- Originally designed by Bosch for interconnection of microcontroller-based devices in vehicles
- Also used in other applications
  - Automated manufacturing
  - Distributed process control
- Standardized as ISO 11898
- Only specifies the physical and data link layers
- Basis for other fieldbus protocols that define their own higher level layers and services
  - CANopen, DeviceNet, ControlNet, Smart Distributed System, etc.

# CAN Message Format




- **Identifier – Identifies the message content and the priority**
  - CAN 2.0B specification allows 29 identification bits
- **Data – Short messages: up to 8 data bytes** (64 data bits)
- DLC (Data Length Code) – Number of data bits
- RTR (Remote Transmission Request) bit – Is  dominant (“1”) when information is required from another node. All nodes receive the request, but the identifier determines the destination node.
- EOF – End of frame, IFS – Interframe space

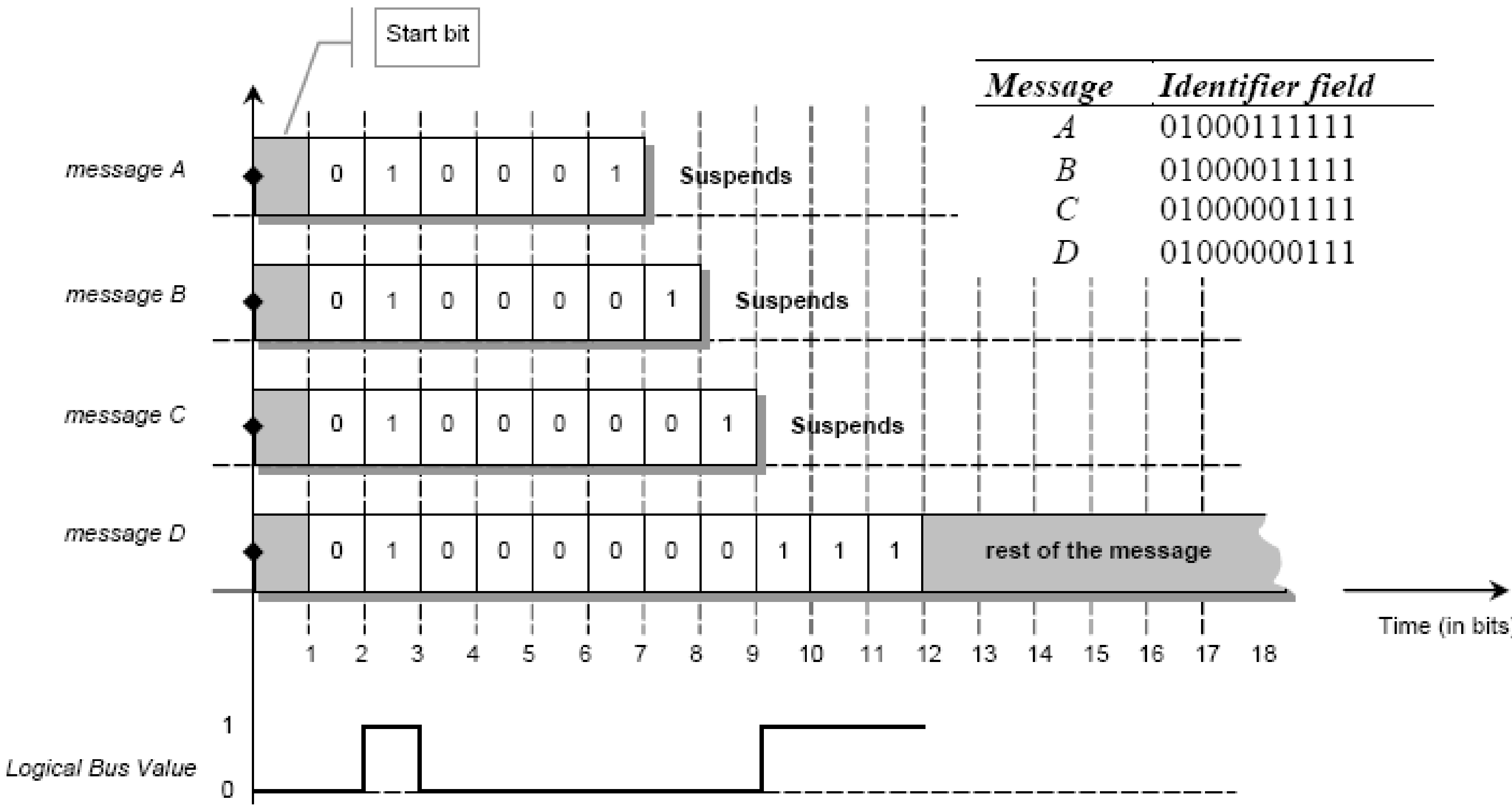
# CAN Protocol

- Distributed medium access coordination
  - Contention based, CSMA-like protocol
  - Priority based
  - **Non-destructive collisions**
    - Unlike typical CSMA protocols
    - **In case of collision, the higher priority message is successfully transmitted**
    - **Collision is detected during the transmission of the identifier field bits**
    - Stations with lower priority messages stop before the transmission of remaining message fields, avoiding the destruction of the message

# CAN Collision Resolution Mechanism

- The open-collector **CAN bus acts as AND-gate**, where each station is able to read the bus status
  - “0” if at least one station transmits a “0”, “1” otherwise
- **When the bus becomes idle, every station with pending messages will start to transmit**
- **During the transmission of the identifier field, if a station is transmitting a “1” and reads a “0”**
  - It means that there was a collision with at least one higher-priority message
  - Consequently, **this station aborts the message transmission**
- **The highest-priority message being transmitted will proceed without perceiving any collision**

# CAN Collision Resolution Example



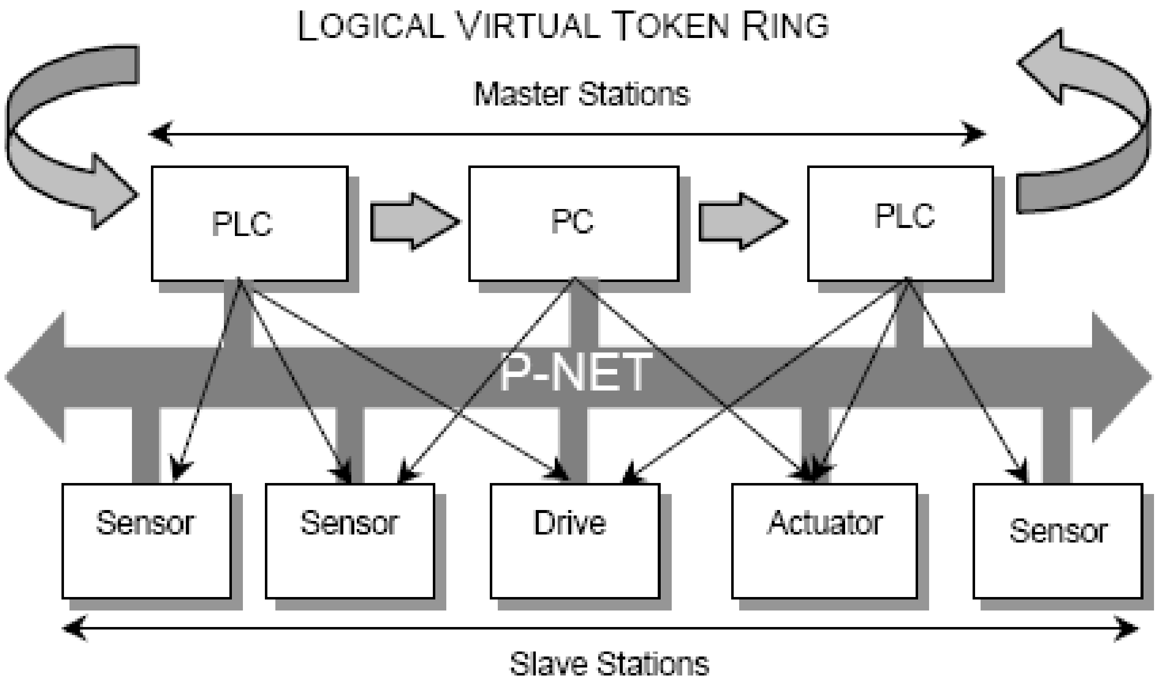

| *Message* | *Identifier field* |
|---|---|
| *A* | 01000111111 |
| *B* | 01000011111 |
| *C* | 01000001111 |
| *D* | 01000000111 |

# CAN PHY Limitations

- CAN collision resolution mechanism imposes
  - **Precise synchronization** among contention stations
    - **At bit level**
- Due to **propagation delay**, this requirement brings a **strict tradeoff between**
  - **Bus length**
  - **Transmission data rate**
- For instance, a bus length of 100 m imposes a maximum data rate of 500 kbps
- Longer buses are only possible at the cost of a data rate reduction, and vice versa

# Process Network (P-NET)

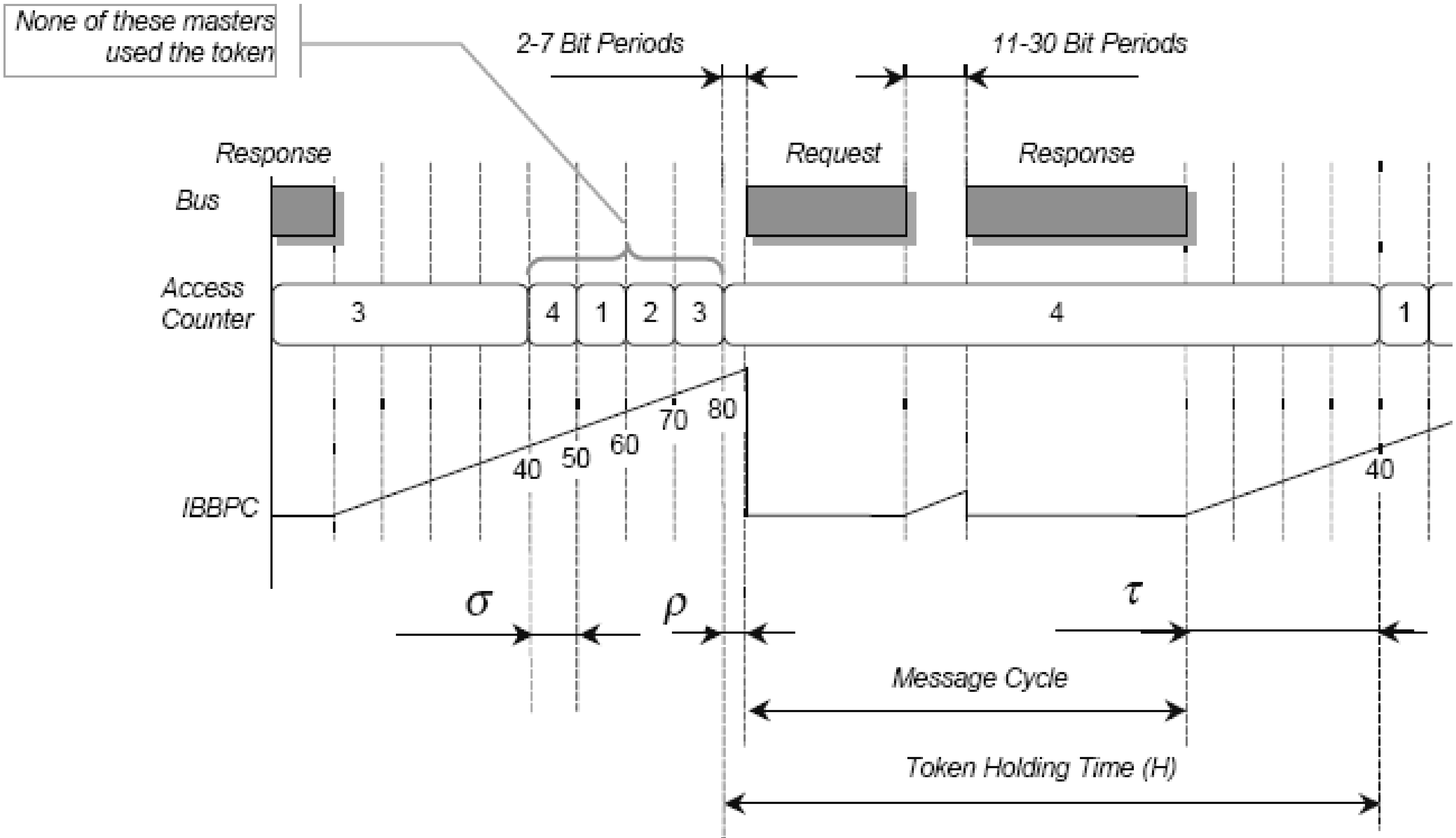
- Data rate of 76.8 kbps
- Multi-master architecture
- Hybrid MAC Protocol
  - Virtual token-passing (VTP) – Among masters
  - Polling – Master-slave communication



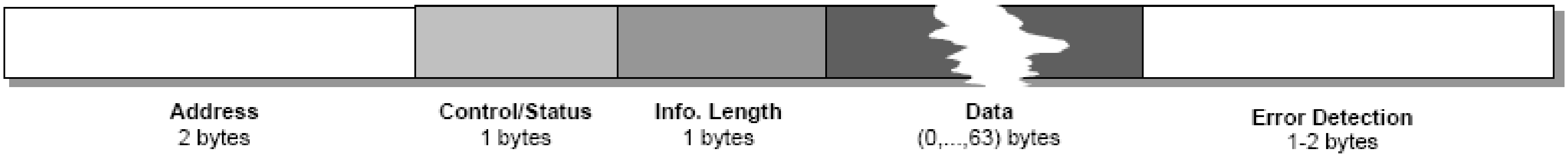
# P-NET VTP Scheme

- Idle bus bit period counter (IBBPC)
  - Increments for each inactive bus bit period
  - Reset to zero when a transmission starts
- Access counter (AC)
  - holds address of currently controlling master
  - Incremented by 1 when the bus has been idle for
    - 40 bit periods after a transmission
    - Further 10 bit periods (50, 60, 70 ...)
- The master whose AC value is equal to its node address
  - Holds the token
  - Controls the access to the bus using polling

# P-NET MAC Protocol Example



# P-NET Message Format




- Each byte is transmitted using 11 bits
  - One start bit (logical zero)
  - 8 data bits, with LSB (Least Significant Bit) first
  - One address/data bit
  - One stop bit
- In a frame, each start bit must follow immediately the previous stop bit
- A frame may have up to 759 bits (69x11 bits)

# PROcess Field BUS (PROFIBUS)

- Multi-master architecture (similar to P-NET)
- Hybrid MAC Protocol
  - Token-passing – Among masters
  - Polling – Master-slave communication
- Four data transmission services
  - **Send Data with No acknowledge** (SDN)
    - **Broadcasts** from a master station to all other stations on the bus
  - **Send Data** with Acknowledge (SDA)
  - **Request Data** with Reply (RDR)
  - **Send and Request Data** (SRD)

# PROFIBUS Message Cycle

- Exchange of two frames
  - Master's action frame (poll frame)
    - Send, request or send/request frame
  - Slave’s response
    - Acknowledgement (ACK) or reply frame
- If the ACK/response does not arrive in a slot time
  - The initiator repeats the request
    - Up to a predefined maximum number of retries, before a communication error report

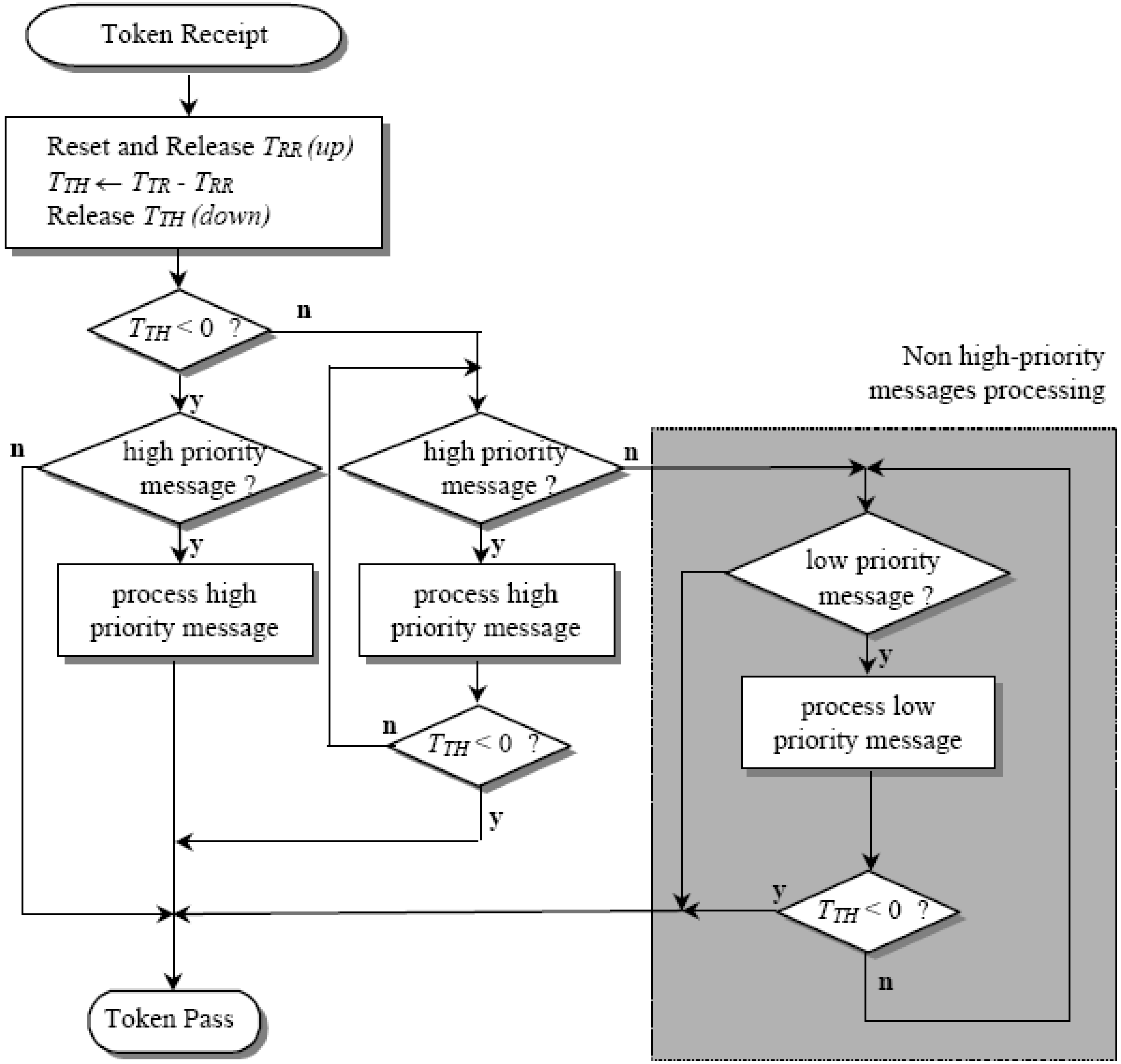
# Control of the Token Cycle Time

- Parameters and variables
  - Target Token Rotation Time ($T_{TR}$) – Expected time to complete a rotation through all the masters
    - Parameter of PROFIBUS network
  - Real Token Rotation Time ($T_{RR}$) – Time elapsed since token was passed the last time
    - Calculated in the beginning of each message cycle
  - Token Holding Time ($T_{TH}$) – Indication of remaining time

$$\boldsymbol{T_{TH} = T_{TR} - T_{RR}}$$

- Main categories of message (separate queues)
  - High priority
  - Low priority

# PROFIBUS Token Cycle Control Algorithm



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